

Keolebogile Gillian Merafe

Exercise of

SQL Fundamentals:

SELECT & Filtering

Selecting & Sorting

① SELECT *
FROM employees;

id	First-name	Last-name	department	Salary	hire-date
1	Alice	Green	IT	70000	2020-01-10
2	Brian	Lee	HR	45000	2019-03-22
3	Cathy	Zulu	Finance	65000	2018-07-18
4	David	Mokoena	Marketing	50000	2021-11-05
5	Eva	Maidoo	IT	72000	2017-09-30

② SELECT DISTINCT department
FROM employees;

Department
IT
HR
FINANCE
MARKETING

③ SELECT first-name,
Last-name

FROM employees
ORDER BY Salary DESC;

First-name	Last-name
Eva	Maidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

④ SELECT Id,
First-name,
Last-name,
Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;

Id	First-name	Last-name	Salary
5	Eva	Maidoo	72 000
1	Alice	Green	70 000
3	Cathy	Zulu	65 000

④ Filtering with WHERE

⑤ SELECT Id,
First-name,
Last-name,
department

FROM employees
WHERE department = 'IT' ;

id	First_name	Last_name	department
1	Alice	Green	IT
5	Eva	Maidoo	IT

⑥ SELECT id,
First_name,
Last_name,
department,
Salary

FROM employees
AND department = 'Finance'
WHERE SALARY > 60 000 ;

id	First_name	Last_name	department	Salary
1	Alice	Green	IT	70 000
3	Cathy	Zulu	FINANCE	65 000
5	Eva	Maidoo	IT	72 000

⑦ SELECT id,
First_name,
Last_name,
department

FROM employees
WHERE department = 'HR' OR 'Marketing' ;

id	First_name	Last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

⑧ SELECT id,
first-name,
last-name,
department

FROM employees
WHERE department = 'IT'

id	First-name	Last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

⑨ SELECT id,
first-name,
last-name,
department

FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

id	First-name	Last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Maidoo	IT

⑩ SELECT id,
first-name,
last-name,
department,
Salary,
City

FROM employees

WHERE department = 'IT'

AND SALARY > 65 000

AND City = 'Johannesburg' ;

id	First_name	Last_name	department	Salary	City
1	Alice	Green	IT	70 000	Johannesburg
3	Cathy	Zulu	Finance	65 000	Durban
5	Eva	Maidoo	IT	72 000	Johannesburg